

ANURAG KUMAR SINGH

Pune, Maharashtra

+91 9004322902 · anurag.k.singh02@gmail.com

[LinkedIN](#), [GITHUB](#), [HackerRank](#)

EDUCATION

Symbiosis Institute of Technology, Pune

2021 - Present

Bachelor of Technology, Artificial Intelligence & Machine Learning

TECHNICAL SKILLS

- **Programming & Scripting:** Python, SQL, JavaScript, Java, HTML/CSS
- **ML & AI Tools:** Scikit-learn, TensorFlow, PyTorch, LLM, Transformers (BERT, RAG), NLP, CV, LangChain
- **Data & Visualization:** Pandas, NumPy, PySpark, Power BI, Tableau, Excel, Matplotlib, Seaborn
- **Cloud & DevOps:** Azure, AWS, GCP, Databricks, ADF, Docker, CI/CD, Git, FastAPI, MLflow, MongoDB
- **Specializations:** Computer Vision, Natural Language Processing (NLP), Transformers, Autoencoders

CERTIFICATIONS *(notable)*

Databricks Certified Data Engineer Associate, **Databricks**

IBM RAG and Agentic AI Professional Certificate, **IBM**

Mastering Data Structures & Algorithms using C and C++, **UDEMY**

EXPERIENCE

R&D Intern (Data Engineering)

Jan 2025 - Jul 2025

DIAC TO Technologies Pvt. Ltd, Pune

Designed and optimized 10+ ETL pipelines using Azure Data Factory and Databricks, processing over 100M records weekly. Executed large-scale data transformations and managed end-to-end workflows, reducing pipeline runtime by 30%. Utilized Python and SQL to build scalable data solutions, improving data availability and processing efficiency across cloud environments

AI Research Intern

May 2024 - Nov 2024

Defence Research and Development Organization (Ministry of Defence, Govt of India) Pune

Built deep generative models (GANs, VAEs) to generate 50K+ novel chemical SMILES. Predicted molecular properties like dipole moments using GNNs (GINs, GCNs) with >85% accuracy. Deployed the pipeline on the cloud, boosting scalability and cutting processing time by 40%, accelerating chemical discovery.

RESEARCH AND PROJECT EXPERIENCE

Security Event Correlation and Threat Detection System (Research Project @ CDAC)

- Developed and compared CNN, LSTM, and Random Forest models for classifying network traffic (CICIDS-2017 dataset), achieving 95% accuracy with CNN.
- Engineered features (byte rate, packet size) for anomaly detection and applied SMOTE for class balancing.
- Integrated SHAP and LIME for model explainability and improved transparency for cybersecurity analysts.

Chest X-Ray Image Generation

- Built a DCGAN model to generate realistic Chest X-ray images, improving dataset diversity by 25% for enhanced medical diagnostics.
- Optimized GAN performance by hyperparameter tuning, reducing training time by 20%.
- Enhanced image resolution to aid in model training and diagnostic tool development.

Road Topology Anomaly Detection

- Designed a real-time pothole detection system using YOLOv8, achieving a 92% accuracy rate in identifying road anomalies.
- Integrated geolocation data for dynamic mapping, enabling rapid maintenance scheduling.
- Streamlined data preprocessing pipelines, reducing latency by 30% in real-time video analysis.

Multimodal Sentiment Analysis using Multitasking

- Implemented an end-to-end NLP pipeline integrating VADER, TF-IDF, Random Forest, NRCLEX, and BERT to classify emotions and predict customer ratings from feedback data.
- Enhanced sentiment analysis accuracy by 20% through advanced text preprocessing and multitask learning, improving model robustness and business insights.

ACHIEVEMENTS AND PARTICIPATIONS

- Ranked in the Top 15 at a project exhibition hosted by DY Patil University.
- Advanced through two rounds in the AI Hackathon for the retirement industry by TIAA.
- Participated in hackathons including HackRx 4.0 (Bajaj), KakushIN 8.0 (Nomura), and many more
- Medalist in Football & Athletics (Team Captain); Cultural Fest Coordinator & Sports Fest Representative.